

Data Flow Diagram

Date	13 July 2026
Team ID	
Project Name	SkillEcho — Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

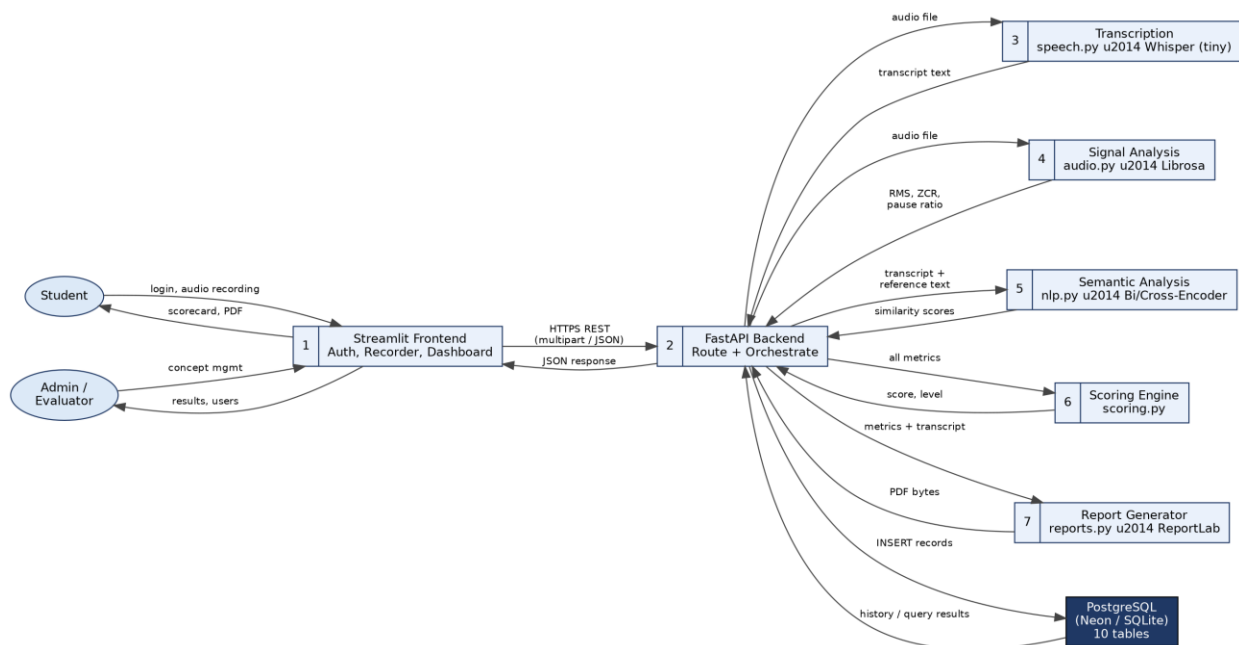
Data Flow Diagram

The diagram below shows how data moves through SkillEcho — between external entities (Student, Admin), the Streamlit/FastAPI processes, the AI service modules, and the PostgreSQL data store. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Student, Admin, Service Provider).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Transcription, Scoring Engine).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., PostgreSQL database, 10 normalised tables).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

SkillEcho — Level 1 Data Flow Diagram



External Data Flow Summary

Source	Data	Destination	Protocol
Student Browser	Audio WAV file	FastAPI /evaluate	HTTPS POST (multipart)
Student Browser	Login credentials	FastAPI /auth/login	HTTPS POST (JSON)
FastAPI	Audio file bytes	data/uploaded_audio/	Local filesystem write
FastAPI	Transcript, metrics	PostgreSQL (Neon Cloud)	SQLAlchemy → psycopg2-binary
FastAPI	Evaluation JSON	Streamlit Frontend	HTTPS JSON response
ReportLab	PDF bytes	Student browser	Streamlit st.download_button()
Admin Browser	Concept text + PDF	FastAPI /admin/concepts	HTTPS POST (multipart)